Since this plane passes through the point $(-1,2,1)$, we get $\lambda = \frac{3}{5}$, and hence the required equation of the plane is $11x-4y+20z=1$.

Example 6.54

Find the equation of the plane passing through the intersection of the planes $2x+3y-z+7=0$ and $x+y-2z+5=0$ and is perpendicular to the plane $x+y-3z-5=0$.

Solution

The equation of the plane passing through the intersection of the planes $2x+3y-z+7=0$ and $x+y-2z+5=0$ is $(2x+3y-z+7)+\lambda(x+y-2z+5)=0$ or

$$(2+\lambda)x+(3+\lambda)y+(-1-2\lambda)z+(7+5\lambda)=0$$

since this plane is perpendicular to the given plane $x+y-3z-5=0$, the normals of these two planes are perpendicular to each other. Therefore, we have

$$(1)(2+\lambda)+(1)(3+\lambda)+(-3)(-1-2\lambda)z=0$$

which implies that $\lambda=-1$.Thus the required equation of the plane is

$$(2x+3y-z+7)-(x+y-2z+5)=0 \Rightarrow x+2y+z+2=0 \ .$$

6.9 Image of a Point in a Plane

Let A be the given point whose position vector is $\vec{u}$. Let $\vec{r}\cdot\vec{n}=p$ be the equation of the plane. Let $\vec{v}$ be the position vector of the mirror image A' of A in the plane. Then $\overrightarrow{AA'}$ is perpendicular to the plane. So it is parallel to $\vec{n}$. Then

$$\overrightarrow{AA'}=\lambda\vec{n} \text{ or } \vec{v}-\vec{u}=\lambda\vec{n} \Rightarrow \vec{v}=\vec{u}+\lambda\vec{n} \qquad \text{... (1)}$$

Let M be the middle point of AA'. Then the position vector of M is $\frac{\vec{u}+\vec{v}}{2}$. But M lies on the plane.

So, $\left(\frac{\vec{u}+\vec{v}}{2}\right)\cdot\vec{n}=p$. ... (2)

Sustituting (1) in (2), we get

$$\left(\frac{\vec{u}+\lambda\vec{n}+\vec{u}}{2}\right)\cdot\vec{n}=p \Rightarrow \lambda=\frac{2\left[p-(\vec{u}\cdot\vec{n})\right]}{|\vec{n}|^2}$$

Therefore, the position vector of A'

is $\vec{v}=\vec{u}+\frac{2[p-(\vec{u}\cdot\vec{n})]}{|\vec{n}|^2}$

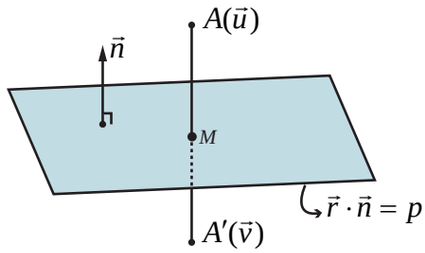


Fig. 6.35

Note

The mid point of M of AA' is the foot of the perpendicular from the point A to the plane $\vec{r}\cdot\vec{n}=p$. So the position vector of the foot M of the perpendicular is given by .

$$\frac{\vec{u}+\vec{v}}{2}=\frac{\vec{u}}{2}+\frac{1}{2}\left(\vec{u}+\frac{2[p-(\vec{u}\cdot\vec{n})}{|\vec{n}|^2}\vec{n}\right)$$

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